

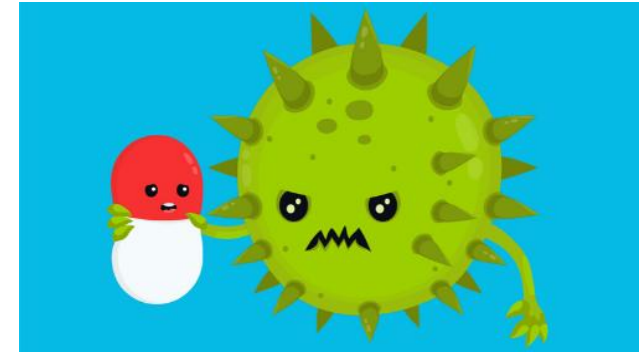


Antimicrobial Drugs Resistance

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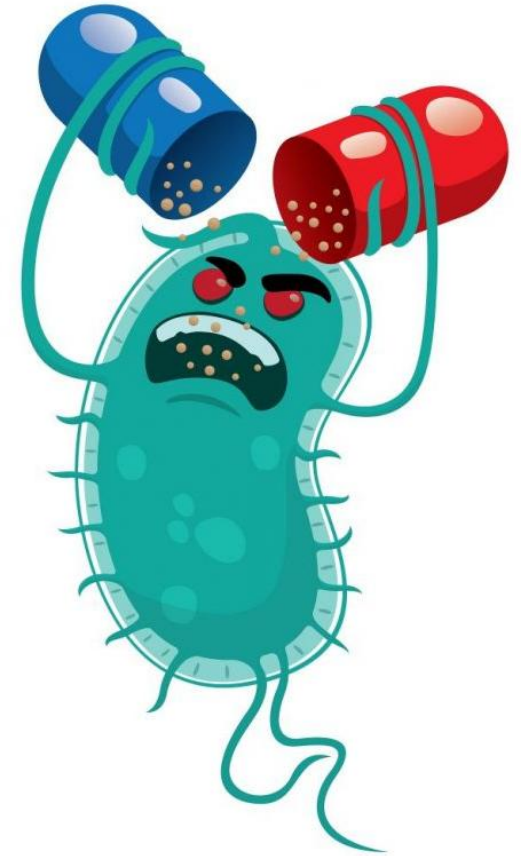
Introduction

- The discovery of **antimicrobials** is one of the great advances in medicine, and their use has substantially reduced **morbidity and mortality** worldwide.
- Unfortunately, with **widespread antibiotic use**, we have witnessed the emergence of **multidrug-resistant pathogens** and reduced efficacy of many of our most powerful antimicrobials.
- A growing number of infections – such as **pneumonia, tuberculosis, gonorrhoea, and salmonellosis** – are becoming harder to treat as the antibiotics used to treat them become less effective.



What are superbugs

- These **antibiotic-resistant bacteria** are called **superbugs**, and they can cause severe bacterial infections that are **extremely difficult to treat**.
- Superbugs are **strains of bacteria, viruses, parasites and fungi** that are resistant to **most of the antibiotics** and other **medications** commonly used to treat the infections they cause.
- Superbug infection symptoms look the same as the symptoms of other infections. The difference is that the **symptoms do not respond** to antibiotics and antifungal medications.



Mechanisms of antibiotic drug resistance among bacteria

There are four major mechanisms that mediate bacterial resistance to drugs :

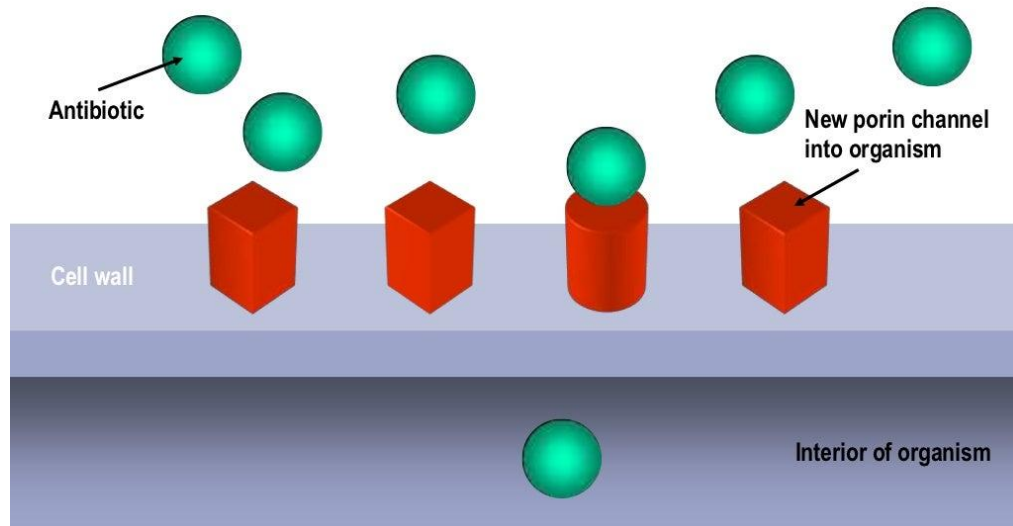
- ❑ **Bacteria produce enzymes that inactivate the drug**, e.g., β -lactamases can inactivate penicillins by cleaving the β -lactam ring of the drug.

β -Lactamase: An enzyme that hydrolyzes the β -lactam ring in the β -lactam class of antibiotics, inactivating the antibiotic. The enzymes specific for penicillins, cephalosporins, and carbapenems are the **penicillinases**, **cephalosporinases**, and **carbapenemases**, respectively.

- ❑ Microorganisms **change their permeability** to the drug.
Examples: Tetracyclines accumulate in susceptible bacteria but not in resistant bacteria.

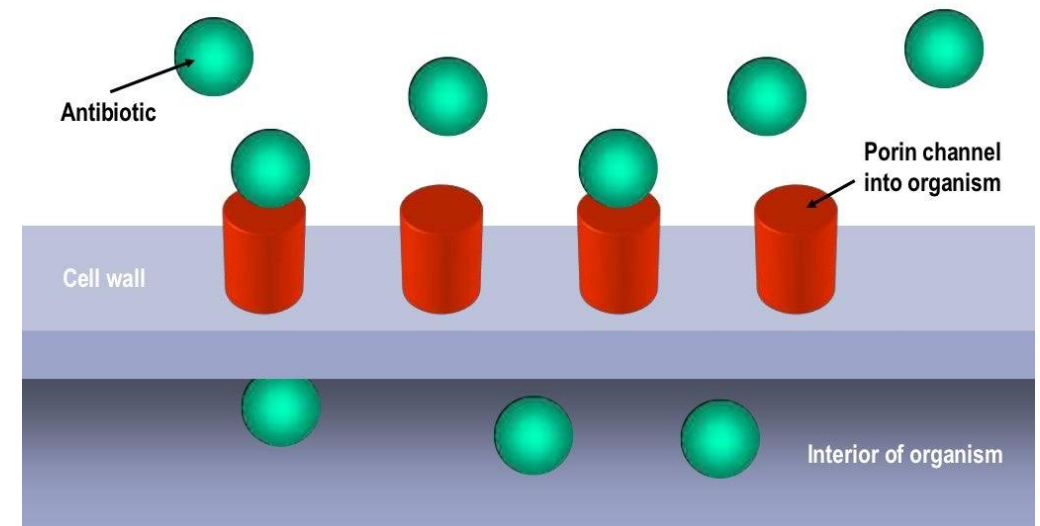
Decreased permeability: Porin Loss

New porin channels in the bacterial cell wall do not allow antibiotics to enter the cells



Decreased permeability: Porin Loss

Antibiotics normally enter bacterial cells via porin channels in the cell wall



❑ Bacteria synthesize modified targets against which the drug has a reduced effect

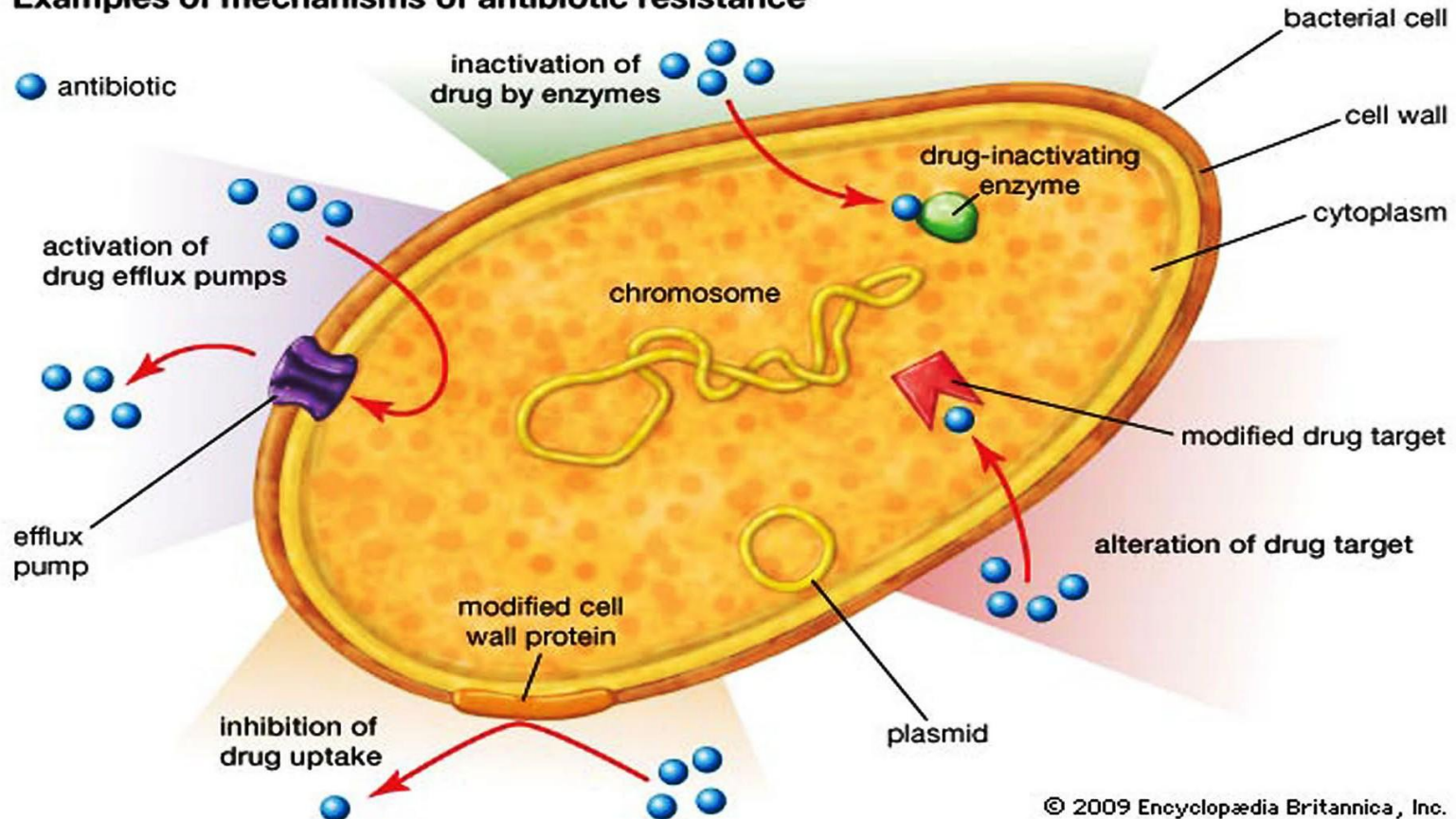
- e.g., a mutant protein in the 30S ribosomal subunit can result in resistance to streptomycin

❑ Bacteria actively export drugs using a “multidrug-resistance pump” (MDR pump).

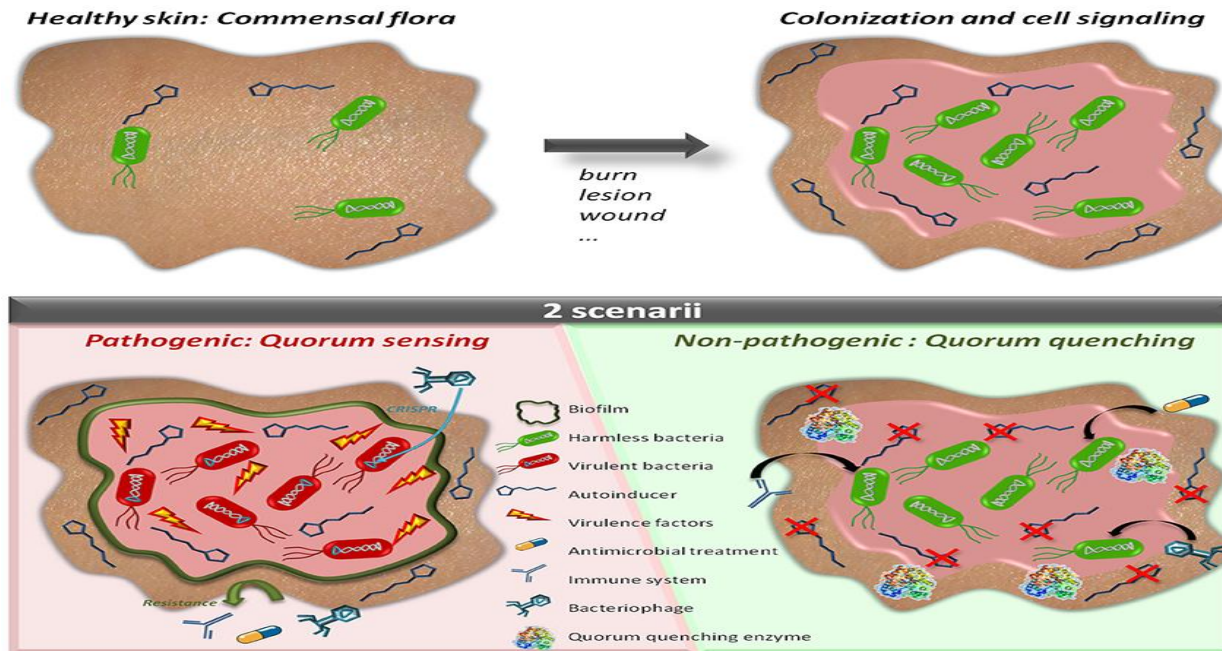
- The MDR pump imports protons and exports a variety of foreign molecules including certain antibiotics, such as tetracyclines.

Mechanisms of antibiotic drug resistance among bacteria

Examples of mechanisms of antibiotic resistance



Quorum sensing



Quorum sensing in a wounded tissue

- Many bacteria exploit a cell-cell communication mechanism called **quorum sensing** to regulate the expression of genes involved in **virulence, competition, pathogenicity** and **resistance**

ORIGIN OF DRUG RESISTANCE

Most drug resistance is due to:

1- Genetic change in the organism

- Chromosomal mutation
- Acquisition of a plasmid or transposon

2- Non-genetic changes, are less importance.

Genetic Basis of Resistance

1 Chromosome-Mediated Resistance

- **Change the target of the drug** so that the drug does not bind
- **Change the membrane** so that the drug does not penetrate well into the cell.
- **Chromosomal mutations occur at a low frequency** (perhaps 1 in 10 million organisms) and often affect only one drug or one family of drugs.
- **Chromosomal resistance is less of a clinical problem than is plasmid mediated resistance.**

2- Plasmid-Mediated Resistance

- **Plasmids cause drug resistance by encoding enzymes that degrade or modify drugs.**
- **Plasmid-mediated resistance occurs at a higher frequency than chromosomal mutations**, often affecting multiple drugs or families of drugs.
- **Resistance plasmids (R plasmids, R factors) usually carry two sets of genes.**
 - Encodes the enzymes that degrade or modify drugs.
 - Encodes the proteins that mediate conjugation, the main process by which resistance genes are transferred from one bacterium to another.

➤ **Plasmid-mediated resistance is very important from a clinical point of view**

(1) Occurs in many different species, especially gram-negative rods.

(2) Frequently mediate resistance to multiple drugs.

(3) Have a high rate of transfer from one cell to another, usually by conjugation.

3- Transposon-Mediated Resistance

Transposons are genes that are transferred either within or between larger pieces of DNA such as the bacterial chromosome and plasmids

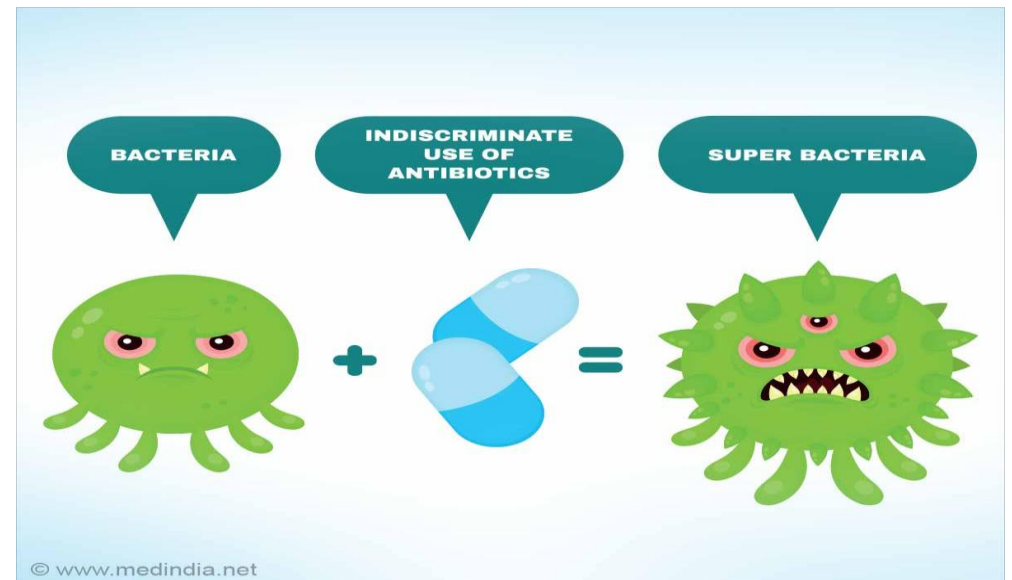
Non-genetic basis of resistance

- There are several nongenetic reasons for the failure of drugs to inhibit the growth of bacteria:
 - 1) **Bacteria can be walled off within an abscess cavity that the drug cannot penetrate effectively.** Surgical drainage is therefore a necessary adjunct to chemotherapy.
 - 2) **Bacteria can be in a resting state (i.e., not growing);** they are therefore insensitive to cell wall inhibitors such as penicillins and cephalosporins.
 - 3) **Under certain circumstances,** organisms that would ordinarily be killed by penicillin can lose their cell walls, survive as protoplasts. Later, if such organisms resynthesize their cell walls, they are fully susceptible to these drugs.

(4) The presence of foreign bodies makes successful antibiotic treatment more difficult. This applies to foreign bodies such as surgical implants and catheters.

(5) Administration of the wrong drug or the wrong dose or failure of the drug to reach the appropriate site in the body

- e.g. the poor penetration into spinal fluid by several early-generation cephalosporins.



Antibiotic resistance is accelerated by

- Misuse of antibiotics
- Overuse of antibiotics,
- Poor infection prevention and control.

Steps can be taken **at all levels of society** to reduce the impact and limit the spread of resistance.

Overuse & Misuse of antibiotics



- Use multiple antibiotics when one would be sufficient
- In many countries, antibiotics are sold over the counter to the general public
- Antibiotics are used in animal feed

➤ CAUSES: Hospital set-up

□Hospitals are ideal reservoirs for the acquisition of resistance

- A population of people with compromised health
- A high concentration of organisms, many of which are extremely pathogenic
- Large and increased amounts of different antibiotics are constantly in use
- Hospital is a place where resistance can develop rapidly

➤ Development of Resistance

- **Bacterial cells that have developed resistance are not killed off.**
 - They continue to divide
 - Resulting in a completely resistant population

➤ Development of Resistance

- These antibiotics do destroy the **normal flora**.
- **Opportunistic** pathogens that are resistant survive and can take hold.

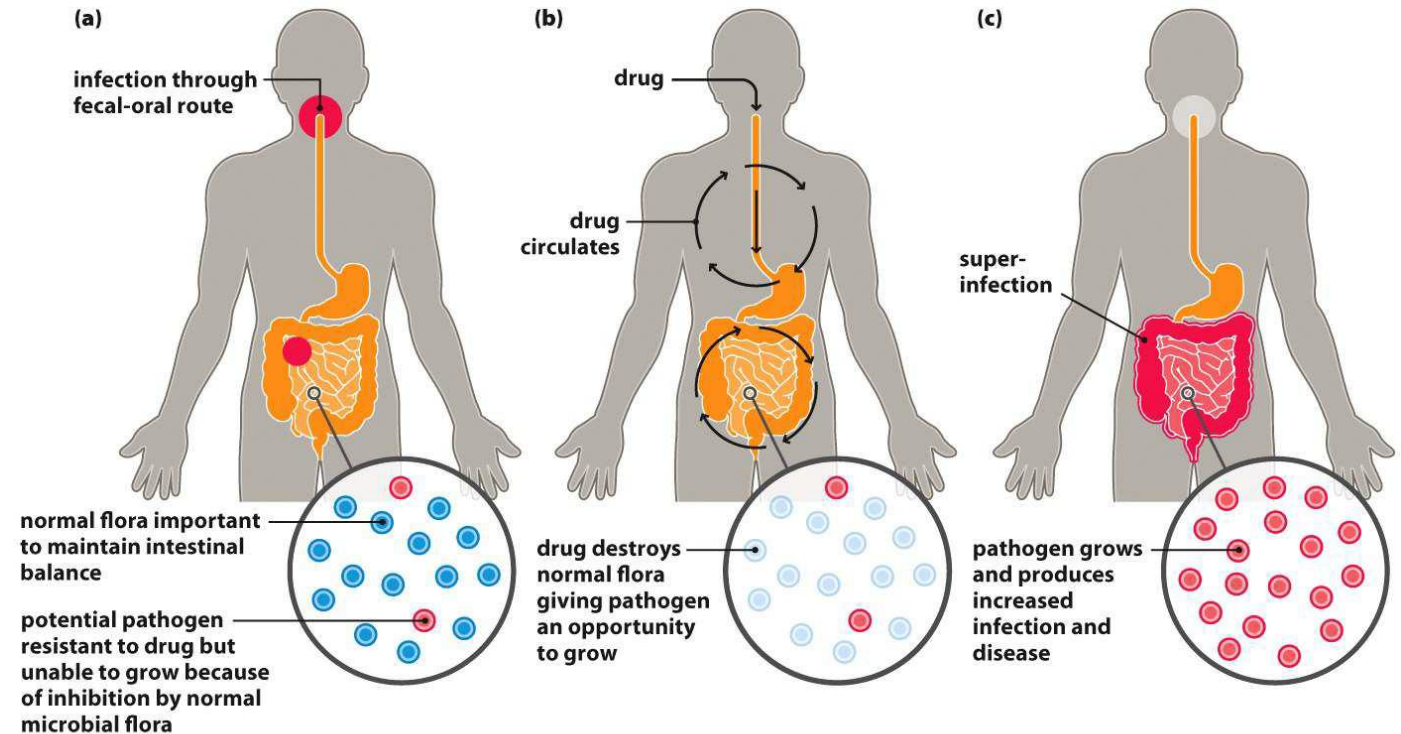


Figure 20.3 Microbiology: A Clinical Approach (© Garland Science)

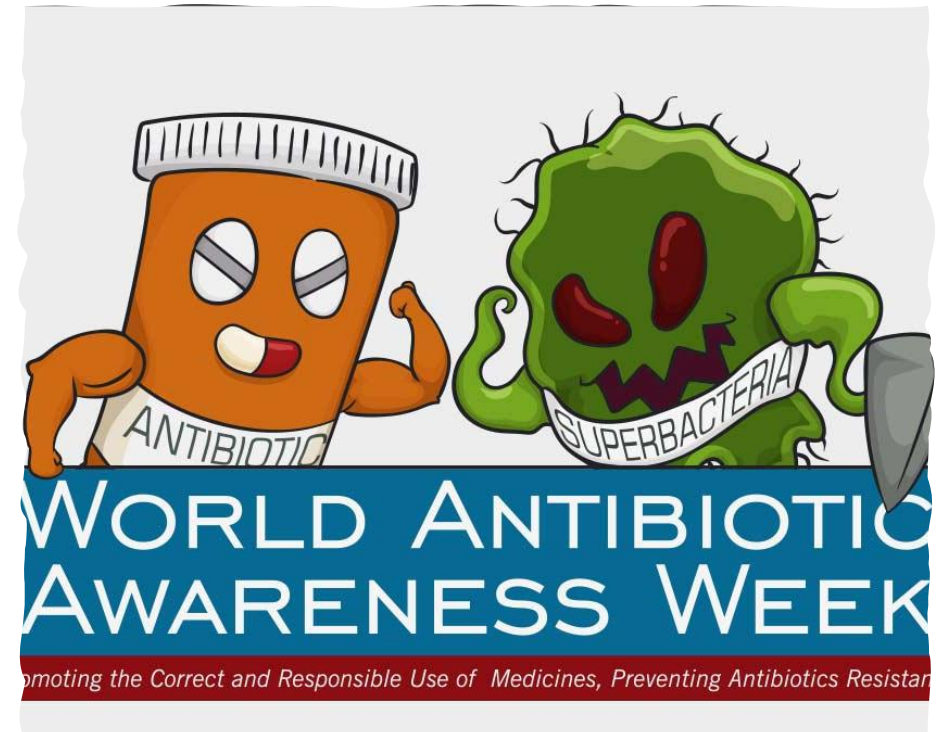
How can we control the spread of MDROs?

- Only use antibiotics when they are needed and as directed
- Washing hands with soap and water (or alcohol hand rub) for at least 20 seconds
- Wear disposable gloves and possibly gowns when applicable
- Family members should be instructed to wash their hands upon entering a patient's room and before leaving the room



World Antimicrobial Awareness Week

- WAAW takes place every year from **18 to 24 November**.
- The slogan has previously been, **“Antibiotics: Handle with Care”** but changed to **“Antimicrobials: Handle with Care”** in 2020 to reflect the broadening scope of drug resistant infections.



Q1/ R factors are _____ that contain a code for _____.

- A. genes, replication
- B. plasmids, drug resistance
- C. transposons, interferon
- D. plasmids, conjugation
- E. plasmids, transduction

Q2/ A compound synthesized by bacteria or fungi that destroys or inhibits the growth of other microbes is a/an

- A. Synthetic drug.
- B. Antibiotic.
- C. Interferon.
- D. Competitive inhibitor
- E. Semi synthetic drug

Q3/ The proportion of antibiotic-resistant bacteria has increased along with the widespread use of antibiotics. This is due to the fact that antibiotics:

- A. Are unstable in vivo
- B. Act as agents of selection for resistant organisms
- C. Are mainly bacteriostatic in vivo
- D. Are powerful mutagens
- E. Are mainly bactericidal in vivo

Q4/ β -Lactamases are an important cause of antibiotic resistance. Which one of the following is the most common site where β -lactamases are located?

- A. Attached to DNA in the nucleoid
- B. Attached to pili on the bacterial surface
- C. Free in the cytoplasm
- D. Within the capsule
- E. Within the periplasmic space

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